Ps 03

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1 Problem 1 DONE

1.1 Task

The ReLU function is a standard activation function in neural networks. It is simply given by $f : \mathbb{R} \to \mathbb{R}$ $f(x) = \begin{cases} 0, & x \leq 0, \\ x, & x > 0. \end{cases}$ Use the (ϵ, δ)-definition of continuity to determine, where f is continuous and where not. Plot f in Julia.

1.2 Notes

- What does it mean "where not"?

  The function f looks to be continuous $\forall x \in \mathbb{R}$. as the $\lim_{x \to 0^+} f(x) = 0$.

  The only suspicion that I have is that it might now be continuous due to $f(x) = x : x > 0$ and not $x \geq 0$, that makes it discontinuous.

1.3 Solution

$$\forall \epsilon > 0 \, \exists \delta > 0 : \forall x, |x - c| < \delta \implies |f(x) - f(c)| < \epsilon$$

The function f is known to be continuous for $\mathbb{R} \backslash 0$, as $\forall x \in \mathbb{R} : x < 0, f(x) = 0$, and $\forall x \in \mathbb{R} : x > 0, f(x) = x$ are known to be continuous. Hence, I will validate f 's continuity on point 0.

Let $\epsilon > 0$. Choose $\delta = |\epsilon|$ such that $|x - 0| < \delta$.

- If $x \leq 0$, then $|f(x) - f(0)| = |0 - 0| = 0 < \epsilon$.
- If $x > 0$, then $|f(x) - f(0)| = x < \delta = \epsilon$.
- Code

  ```
  nil
  ```

2 Problem 2 DONE

2.1 Task

Define another activation function $g : \mathbb{R} \to \mathbb{R}$,

$$g(x) = \log(1 + e^x)$$

(a) Why is g continuous on $\mathbb{R}$? (b) Plot f and g in Julia. Which activation function looks "smoother", f or g?

2.2 Solution

- Part (a)

$$g(x) = \log(1 + e^x)$$

Since $h(x) = 1 + e^x$ is known to be continuous $\forall x \in \mathbb{R}$, and $r(x) = \log(x)$ is known to be continuous $\forall x \in (0, \infty)$, the function $g(x) = r \circ h = \log(1 + e^x)$ is a composite of two continuous functions and hence is also continuous $\forall x \in \mathbb{R}$ (as $1 + e^x \geq 1 > 0$).

- Part (b)

```
using Plots

f(x) = (x <= 0) ? 0 : x
g(x) = log(1 + exp(x))

x = -5.0:0.01:5.0
f_x = f.(x)
g_x = g.(x)

plot(x, f_x,
     title="Activation Functions",
     label="f(x)",
     xlabel="x", ylabel="y",
     lw=2)

plot!(x, g_x,
      label="g(x)",
      lw=2)

savefig("./figs/p_02.png")
```

```
/Users/ziro2173/personal/university/01_bachelor/2025_winter_term/01_analysis/02_exercises/s
```

The activation function $g(x) = \log(1 + e^x)$ is looking smoother.

3 Problem 3

3.1 Task

A data set contains the locations $X = (1, 2, 3, 4, 5, 6, 7, 8, 9)^\top$ with corresponding observations $Y = (3, 5, 7, 7, 11, 15, 19, ?, 27)^\top$ The observation at $x = 8$ is missing. (a) Provide an educated guess for the observation at $x = 8$. (b) Write new or modify existing Julia code to fill in missing observations.

3.2 Notes

3.3 Solution

```
nil
```

4 Problem 4 DONE

4.1 Task

We want to estimate the value of exp(1.3) by the following procedure:

1. Use Julia to implement a finite sum to determine $e = \exp(1)$ up to an error smaller than 10^{-6} according to Proposition 1.3.12.
2. Express exp(1.3) by using exp(1) and exp(0.3).
3. Replace the "unknown" value of exp(0.3) by a linear approximation from exp(0) to exp(1).
4. How big is the error?

4.2 Solution

- Part (1)

$$f(x) = e^x$$

Term	Derivative	Value at $x_0 = 0$	Coefficient $\frac{f^{(k)}(0)}{k!}$
$k = 0$	e^x	1	$\frac{1}{1}$
$k = 1$	e^x	1	$\frac{1}{1}$
$k = 2$	e^x	1	$\frac{1}{2!}$
$k = 3$	e^x	1	$\frac{1}{3!}$
...	...	...	...

$$\begin{aligned} e^x &= \sum_{k=0}^{\infty} \frac{x^k}{k!} \\ e^1 &= \frac{1}{1} + \frac{1}{1} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + ... \\ &= \sum_{k=0}^{\infty} \frac{1}{k!} \end{aligned}$$

```
function approx_e(tol=1e-6)
    k = 1
    sum = 0.0
    term = 1.0

    while term >= tol
        sum += term
        term *= 1/k
        k += 1
    end

    return sum
end

approx_e()
```

```
2.7182815255731922
```

- Part (2)

  Observe that:

$$e^{1.3} = e^1 \cdot e^{0.3} \qquad \text{(basic exponent property)}$$

- Part (3)

  Knowing the values of both e^0 and e^1 (1, and $2.718\ldots$ respectfully), the value $e^{0.3}$ can be approxiamted by using *Linear Interpolation method*.

$$y_2 = y_1 + (x_2 - x_1)\frac{y_3 - y_1}{x_3 - x_1}$$

```
function approx_e(tol=1e-6)
    k = 1
    sum = 0.0
    term = 1.0

    while term >= tol
        sum += term
        term *= 1/k
        k += 1
    end
```

```
    return sum
end

function approx_e_power(p) # within e^0 and e^1
    x_1, y_1 = 0.0, 1.0
    x_2 = p
    x_3, y_3 = 1.0, approx_e()

    return y_1 + (x_2 - x_1)*(y_3 - y_1)/(x_3 - x_1)
end

approx_e_power(0.3)
```

```
1.5154844576719575
```

- Part (4)

```
function approx_e(tol=1e-6)
    k = 1
    result = 0.0
    term = 1.0

    while term >= tol
        result += term
        term *= 1/k
        k += 1
    end
    return result
end

function approx_e_power(p)
    x_start, y_start = 0.0, 1.0        # exp(0)
    x_end,   y_end   = 1.0, approx_e() # exp(1)

    slope = (y_end - y_start) / (x_end - x_start)

    return y_start + (p - x_start) * slope
end

val_e1 = approx_e()             # exp(1)
val_e03 = approx_e_power(0.3)   # exp(0.3) approx

est_1_3 = val_e1 * val_e03

true_val = exp(1.3)
error = abs(true_val - est_1_3)
```

```
println("Estimated exp(1.3): $est_1_3")
println("True exp(1.3):      $true_val")
println("Absolute Error:     $error")
```

```
approx_e (generic function with 2 methods)
approx_e_power (generic function with 1 method)
2.7182815255731922
1.5154844576719575
4.119513403582991
3.6692966676192444
0.4502167359637461
Estimated exp(1.3): 4.119513403582991
True exp(1.3):      3.6692966676192444
Absolute Error:     0.4502167359637461
```

5 Problem 5

5.1 Task

Consider the two functions $f, g : (0, \infty) \to \mathbb{R}$ given by

$$f(x) = \log(x), \quad g(x) = 2x^2 - 2$$

Plot both functions in Julia and provide mathematical reasons for the number of intersection points that you may observe.

5.2 Notes

5.3 Solution

```
nil
```

6 Problem 6 DONE

6.1 Task

Write Julia functions approxSin and approxCos that approximate sin and cos by using their truncated series expansions. Plot both functions using your approximations.

6.2 Notes

$$T(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!} \cdot (x - x_0)^k$$

$x_0 = 0$

$$\begin{aligned}
T_{cos}(x) &= \frac{\cos(0)}{1} \cdot x^0 + \frac{-\sin(0)}{1} \cdot x^1 + \frac{-\cos(0)}{2} \cdot x^2 + \frac{\sin(0)}{6} \cdot x^3 + ... \\
&= 1 - \frac{1}{2}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + ... \\
&= \sum_{k=0}^{n} \frac{(-1)^k}{(2k)!} x^{2k} \\
\cos(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} \\
\sin(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}
\end{aligned}$$

6.3 Solution

```
using Plots

taylor_cos(x) = sum(((-1)^k * x^(2k))/(factorial(2k)) for k in 0:10)
taylor_sin(x) = sum(((-1)^k * x^(2k+1))/(factorial(2k+1)) for k in 0:9)

x = -pi:0.1:pi

plot(x, taylor_cos.(x), label="Taylor Cos", lw=2)

# 3. Compare it to the REAL cosine (optional but recommended)
plot!(x, taylor_sin.(x), label="Taylor Sin", lw=2)
savefig("./figs/p_06.png")
```

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